

CSE 160 (Fall 2020) – Homework 2

Due: Oct 12 at 11:59pm PT on Canvas.

Total: 10 points.

Modeling

1. Define uniform, attribute and varying variables.

Solution. *Uniform variables are constant to a primitive variable. Attribute variables are per vertex variables that can be positions or colors. And finally Varying variables are per fragment/pixel parameters.*

2. What type is the shader and describe its function?

(a)

```
attribute vec2 a_Position;
main() {
    gl_Position = vec4(0.5 * a_Position, 0.0, 1.0);
}
```

Solution. *Vertex shader - sets a position half of the 2D position passed in a 4D space.*

(b)

```
main() {
    gl_FragColor = vec4(0.0, 0.0, 1.0, 1.0);
}
```

Solution. *Fragment Shader - sets a color to the specified RGBA value divided by 255.*

3. Using linear interpolation, calculate the color of the following points that are on the line defined by the 2D points $p_1 = [3, 5]$ with color $[255, 0, 0]$ and $p_2 = [10, 5]$ with color $[0, 0, 255]$

(a) $[4.8, 5]$

Solution. *Linear Interpolation Eq:*

$$y = y_0 + (x - x_0) \cdot \frac{y_1 - y_0}{x_1 - x_0}$$

For R in $[R, G, B]$:

$$y = 255 + (4.8 - 3) \cdot \frac{0 - 255}{10 - 3} \approx 189.429 = 189$$

For B in $[R, G, B]$:

$$y = 0 + (4.8 - 3) \cdot \frac{255 - 0}{10 - 3} \approx 65.5714 = 66$$

Final Color: $[189, 0, 66]$

(b) $[7.5, 5]$

Solution. For R in $[R, G, B]$:

$$y = 255 + (7.5 - 3) \cdot \frac{0 - 255}{10 - 3} \approx 91.0714 = 91$$

For B in $[R, G, B]$:

$$y = 0 + (7.5 - 3) \cdot \frac{255 - 0}{10 - 3} \approx 163.929 = 164$$

Final Color: $[91, 0, 164]$

(c) $[9, 5]$

Solution. For R in $[R, G, B]$:

$$y = 255 + (9 - 3) \cdot \frac{0 - 255}{10 - 3} \approx \frac{255}{7} = 36$$

For B in $[R, G, B]$:

$$y = 0 + (9 - 3) \cdot \frac{255 - 0}{10 - 3} \approx \frac{1530}{7} = 219$$

Final Color: $[36, 0, 219]$

4. Using barycentric coordinates, calculate the color for each of the following points inside the triangle defined by points $p_1 = [0, 0, 0]$, $p_2 = [3, 5, 0]$ and $p_3 = [6, 0, 0]$ with respective RGB colors $c_1 = [255, 0, 0]$, $c_2 = [0, 255, 0]$, $c_3 = [0, 0, 255]$

(a) $[1, 1, 0]$

Solution. *Total Area of Triangle:*

$$\frac{1}{2} \cdot \|([3, 5, 0] - [0, 0, 0]) \times ([6, 0, 0] - [0, 0, 0])\| = 15$$

$$v_1 = p_1 - [1, 1, 0] = [-1, -1, 0]$$

$$v_2 = p_2 - [1, 1, 0] = [2, 4, 0]$$

$$v_3 = p_3 - [1, 1, 0] = [5, -1, 0]$$

$$A_{c_1 c_2} = \frac{\frac{1}{2} \cdot \|v_1 \times v_2\|}{15} = \frac{1}{15}$$

$$A_{c_1 c_3} = \frac{\frac{1}{2} \cdot \|v_1 \times v_3\|}{15} = \frac{3}{15}$$

$$A_{c_3 c_2} = \frac{\frac{1}{2} \cdot \|v_3 \times v_2\|}{15} = \frac{11}{15}$$

The color at the point $[1, 1, 0]$ is:

$$A_{c_3 c_2} \cdot c_1 + A_{c_1 c_3} \cdot c_2 + A_{c_1 c_2} \cdot c_3 =$$

$$\frac{11}{15} \cdot [255, 0, 0] + \frac{3}{15} \cdot [0, 255, 0] + \frac{1}{15} \cdot [0, 0, 255] = [181, 51, 17]$$

(b) $[3, 4, 0]$

Solution. *Total Area of Triangle:*

$$\frac{1}{2} \cdot ||([3, 5, 0] - [0, 0, 0]) \times ([6, 0, 0] - [0, 0, 0])|| = 15$$

$$v_1 = p_1 - [3, 4, 0] = [-3, -4, 0]$$

$$v_2 = p_2 - [3, 4, 0] = [0, 1, 0]$$

$$v_3 = p_3 - [3, 4, 0] = [3, -4, 0]$$

$$A_{c_1 c_2} = \frac{\frac{1}{2} \cdot ||v_1 \times v_2||}{15} = \frac{\frac{3}{2}}{15}$$

$$A_{c_1 c_3} = \frac{\frac{1}{2} \cdot ||v_1 \times v_3||}{15} = \frac{12}{15}$$

$$A_{c_3 c_2} = \frac{\frac{1}{2} \cdot ||v_3 \times v_2||}{15} = \frac{\frac{3}{2}}{15}$$

The color at the point $[1, 1, 0]$ is:

$$A_{c_3 c_2} \cdot c_1 + A_{c_1 c_3} \cdot c_2 + A_{c_1 c_2} \cdot c_3 =$$

$$\frac{\frac{3}{2}}{15} \cdot [255, 0, 0] + \frac{12}{15} \cdot [0, 255, 0] + \frac{\frac{3}{2}}{15} \cdot [0, 0, 255] = [25.5, 204, 25.5] = [26, 204, 26]$$

(c) $[5, 0.25, 0]$.

Solution. *Total Area of Triangle:*

$$\frac{1}{2} \cdot ||([3, 5, 0] - [0, 0, 0]) \times ([6, 0, 0] - [0, 0, 0])|| = 15$$

$$v_1 = p_1 - [5, 0.25, 0] = [-5, -0.25, 0]$$

$$v_2 = p_2 - [5, 0.25, 0] = [-2, 4.75, 0]$$

$$v_3 = p_3 - [5, 0.25, 0] = [1, -0.25, 0]$$

$$A_{c_1 c_2} = \frac{\frac{1}{2} \cdot ||v_1 \times v_2||}{15} = \frac{12.125}{15}$$

$$A_{c_1 c_3} = \frac{\frac{1}{2} \cdot ||v_1 \times v_3||}{15} = \frac{0.75}{15}$$

$$A_{c_3 c_2} = \frac{\frac{1}{2} \cdot ||v_3 \times v_2||}{15} = \frac{2.125}{15}$$

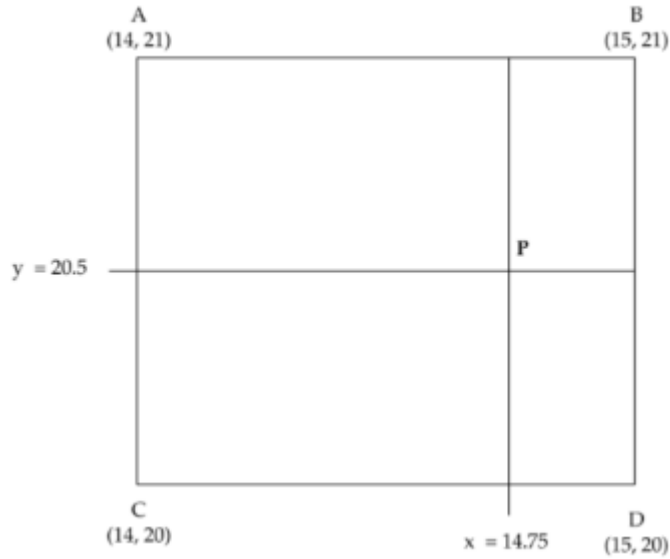
The color at the point $[1, 1, 0]$ is:

$$A_{c_3 c_2} \cdot c_1 + A_{c_1 c_3} \cdot c_2 + A_{c_1 c_2} \cdot c_3 =$$

$$\frac{2.125}{15} \cdot [255, 0, 0] + \frac{0.75}{15} \cdot [0, 255, 0] + \frac{12.125}{15} \cdot [0, 0, 255] = [36.125, 12.75, 206.125] = [36, 13, 206]$$

5. Calculate the RGB color value for point P in the diagram below using bilinear interpolation. Show the intermediate values that you compute for interpolating along one of the two axes. The colors of the points A, B, C and D are the following:

$$A = \text{RGB}(0, 0, 0) \quad B = \text{RGB}(200, 200, 200) \quad C = \text{RGB}(200, 0, 0) \quad D = \text{RGB}(200, 0, 0)$$



Solution. *Bilinear Interpolation formula:*

$$\begin{aligned}
 P &= \frac{(x_2 - x)(y_2 - y)}{(x_2 - x_1)(y_2 - y_1)} \cdot Q_{11} + \frac{(x - x_1)(y_2 - y)}{(x_2 - x_1)(y_2 - y_1)} \cdot Q_{21} \\
 &\quad + \frac{(x_2 - x)(y - y_1)}{(x_2 - x_1)(y_2 - y_1)} \cdot Q_{12} + \frac{(x - x_1)(y - y_1)}{(x_2 - x_1)(y_2 - y_1)} \cdot Q_{22} \\
 P &= \frac{(15 - 14.75)(20 - 20.5)}{(15 - 14)(20 - 21)} \cdot [0, 0, 0] + \frac{(14.75 - 14)(20 - 20.5)}{(15 - 14)(20 - 21)} \cdot [200, 200, 200] \\
 &\quad + \frac{(15 - 14.75)(20.5 - 21)}{(15 - 14)(20 - 21)} \cdot [200, 0, 0] + \frac{(14.75 - 14)(20.5 - 21)}{(15 - 14)(20 - 21)} \cdot [200, 0, 0] \\
 P &= [0, 0, 0] + [75, 75, 75] + [50, 0, 0] + [50, 0, 0] = [175, 75, 75]
 \end{aligned}$$

6. Given a polygon with vertices P_1, P_2, P_3 , each represented by a 3-vector $[x \ y \ z]$, what formula calculates the face's normal?

Solution.

$$\begin{aligned}
 A &= [P_{2x} - P_{1x}, P_{2y} - P_{1y}] \\
 B &= [P_{3x} - P_{1x}, P_{3y} - P_{1y}] \\
 N_x &= A_y * B_z - A_z * B_y \\
 N_y &= A_z * B_x - A_x * B_z \\
 N_z &= A_x * B_y - A_y * B_x \\
 N &= \frac{[N_x, N_y, N_z]}{||[N_x, N_y, N_z]||}
 \end{aligned}$$

Transformations

1. Using the labels Translate, Rotation, Scale, or Reflect, label the transformation of point p when multiplied by matrix M ($p' = Mp$):

(a)

$$M = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Solution. M is a Rotation Matrix

(b)

$$M = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$$

Solution. M is a Rotation Matrix

(c)

$$M = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Solution. M is a Scale Matrix

(d)

$$M = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Solution. M is a Reflect Matrix

(e)

$$M = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

Solution. M is a Translation Matrix

2. Find the matrix for a 120° rotation about the axis defined by the vector $r = (1, 1, 0)$.

Solution.

$$\begin{aligned} R &= \begin{bmatrix} r_x^2 \cdot (1 - \cos \theta) + \cos \theta & r_x \cdot r_y \cdot (1 - \cos \theta) - r_z \cdot \sin \theta & r_x \cdot r_z \cdot (1 - \cos \theta) + r_y \cdot \sin \theta & 0 \\ r_x \cdot r_y \cdot (1 - \cos \theta) + r_z \cdot \sin \theta & r_y^2 \cdot (1 - \cos \theta) + \cos \theta & r_z \cdot r_y \cdot (1 - \cos \theta) - r_x \cdot \sin \theta & 0 \\ r_x \cdot r_z \cdot (1 - \cos \theta) - r_y \cdot \sin \theta & r_z \cdot r_y \cdot (1 - \cos \theta) + r_x \cdot \sin \theta & r_z^2 \cdot (1 - \cos \theta) + \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1^2 \cdot (1 - \cos 120) + \cos 120 & 1 \cdot 1 \cdot (1 - \cos 120) - 0 \cdot \sin 120 & 1 \cdot 0 \cdot (1 - \cos 120) + 1 \cdot \sin 120 & 0 \\ 1 \cdot 1 \cdot (1 - \cos 120) + 0 \cdot \sin 120 & 1^2 \cdot (1 - \cos 120) + \cos 120 & 0 \cdot 1 \cdot (1 - \cos 120) - 1 \cdot \sin 120 & 0 \\ 1 \cdot 0 \cdot (1 - \cos 120) - 1 \cdot \sin 120 & 0 \cdot 1 \cdot (1 - \cos 120) + 1 \cdot \sin 120 & 0^2 \cdot (1 - \cos 120) + \cos 120 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} 1 & \frac{3}{2} & \frac{\sqrt{3}}{2} & 0 \\ \frac{3}{2} & 1 & -\frac{\sqrt{3}}{2} & 0 \\ -\frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

3. Match the following 2D homogeneous matrices to the transformations in the image:

(a)

$$M = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{\sqrt{1}}{2} & 0 \\ \frac{\sqrt{1}}{2} & -\frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Solution. *Graph 2*

(b)

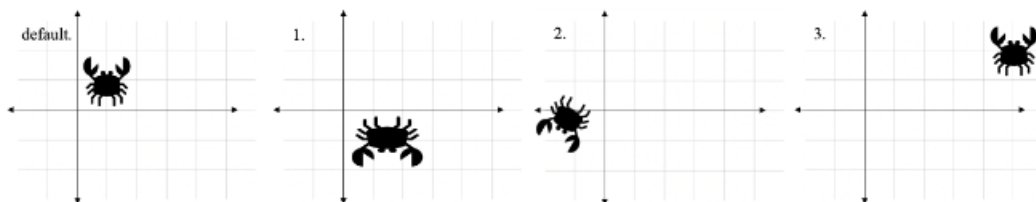
$$M = \begin{bmatrix} 1.5 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Solution. *Graph 1*

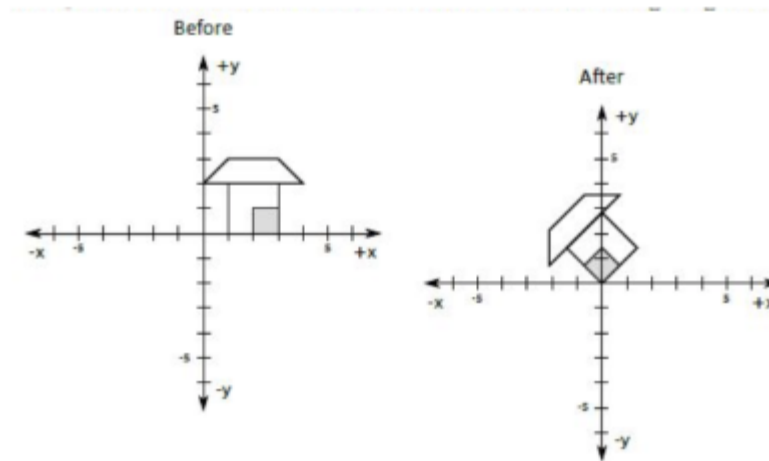
(c)

$$M = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Solution. *Graph 3*



4. Describe a sequence of Translate(x,y), Rotate(degrees), and/or Scale (by a value of) that when multiplied describe the below transformation to go from the 'Before' to 'After'.

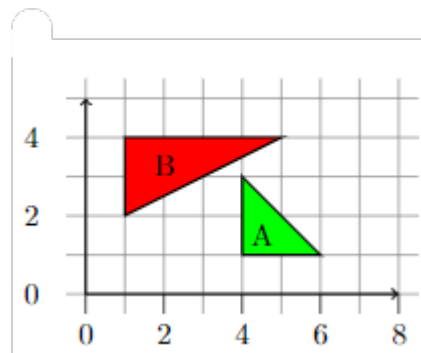


Solution. In order:

- (a) *Translate* $(-2,0)$
- (b) *Scale* $(-1,1)$ [*Reflect across y axis*]
- (c) *Rotate* (90)

5. Describe (using a sequence of (3×3) matrix multiplications) the transformations needed to transform the triangle from A to B in the figure below:

(a)



Solution. In order:

- i. *Translate* $(-4,-1)$
- ii. *Scale* $(1,2)$
- iii. *Rotate* (-90)
- iv. *Translate* $(1,4)$

$$T_1 = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$S = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T_2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

Transformation matrix:

$$T_1 \cdot S \cdot R \cdot T_2 \cdot p = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \cdot p$$

$$T_1 \cdot S \cdot R \cdot T_2 \cdot p = \begin{bmatrix} 0 & 2 & -1 \\ -1 & 0 & 8 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) Using the transformations found in part (a), multiply the following points with the matrix.

i. $P_1 = (4, 1)$

Solution.

$$TMat = \begin{bmatrix} 0 & 2 & -1 \\ -1 & 0 & 8 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix} = (1, 4)$$

ii. $P_2 = (4, 3)$

Solution.

$$TMat = \begin{bmatrix} 0 & 2 & -1 \\ -1 & 0 & 8 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ 1 \end{bmatrix} = (5, 4)$$

Readings Ch3 & Ch4

1. Given a point $p(x, y, z)$ and the point $p'(x', y', z')$ where $p'(x', y', z')$ is the point $p(x, y, z)$ translated by T_x, T_y, T_z :

(a) Show the equations to compute p' from p .

Solution.

$$x' = x + T_x$$

$$y' = y + T_y$$

$$z' = z + T_z$$

(b) Show the 4×4 transformation matrix to compute p' from p .

Solution.

$$\begin{bmatrix} 1 & 0 & 0 & T_x \\ 0 & 1 & 0 & T_y \\ 0 & 0 & 1 & T_z \\ 1 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix}$$

2. Given a point $p(x, y, z)$ and the point $p'(x', y', z')$ where $p'(x', y', z')$ is the β degree rotated point of $p(x, y, z)$ around the z-axis:

- (a) Show the equations to compute p' from p .

Solution.

$$x' = x \cos \beta - y \sin \beta$$

$$y' = x \sin \beta + y \cos \beta$$

$$z' = z$$

- (b) Show the 4×4 transformation matrix to compute p' from p .

Solution.

$$\begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix}$$

3. Given a point $p(x, y, z)$ and the point $p'(x', y', z')$ where $p'(x', y', z')$ is the point $p(x, y, z)$ scaled by S_x, S_y, S_z :

- (a) Show the equations to compute p' from p .

Solution.

$$x' = S_x \cdot x$$

$$y' = S_y \cdot y$$

$$z' = S_z \cdot z$$

- (b) Show the 4×4 transformation matrix to compute p' from p .

Solution.

$$\begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix}$$

4. What is a model matrix? Write a simple vertex shader that uses a model matrix. Explain your shader.

Solution.

```
var VSHADER_SOURCE =
    `attribute vec4 a_Position;
    uniform mat4 u_ModelMatrix;
    void main() {
        gl_Position = u_ModelMatrix * a_Position;
    }`
```

The vertex shader here takes the position attribute of the shape and multiplies it with a Matrix that will transform the points accordingly. The model matrix is essentially the way to apply transformations to points in WebGL. EZ Matrix Multiplication